

HARSH TRIPATHI

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EDUCATION

Indian Institute of Technology (IIT) Bhilai

Bachelor of Technology in Electrical Engineering

2023 – 2027

Chhattisgarh, India

EXPERIENCE

Software Engineer Intern (Backend)

Swir AI

2026 – Present

Remote

- Building the backend for an AI **agentic browser** that autonomously performs end-to-end **web application testing (QA)** for companies, replacing manually written test scripts.
- Developing scalable, event-driven backend services and APIs that drive an **LLM agent** to navigate, interact with, and validate web UIs, then report failures with reproducible traces.

Research Intern

Defence Research and Development Organisation (DRDO)

May 2025 – July 2025

India

- Built **Lunor**, a fully offline, cross-platform neural machine translation desktop app (Rust/Tauri front end + Python/FastAPI backend) serving Meta's **NLLB-200** model.
- Achieved a **BLEU score of 25.02** on Burmese→English, **2.7×** Google Translate's 9.14 baseline; benchmarked MarianMT, Argos, and LibreTranslate across Hindi, Chinese, and Burmese.

Machine Learning Engineer — Explainable AI Trading System

Inter-IIT Tech Meet

Nov 2025 – Dec 2025

India

- Built a real-time algorithmic trading platform using **PPO reinforcement learning**, achieving a **22.3% return** and ranking **12th among 23 IITs**.
- Engineered a streaming data pipeline with **Pathway** and **Redis Streams**, and a multi-agent LLM **Bull-vs-Bear** debate for explainable trade decisions.

PROJECTS

VERIFAI — Verified Evidence-Based Radiology AI | *Python, PyTorch, LangGraph, MedGemma-4B, Next.js*

[GitHub](#)

- Engineered a **9-agent** chest X-ray diagnosis system orchestrated on a **LangGraph** state machine, with safety guardrails, adversarial agent debate, and human-in-the-loop review.
- Designed a **Monotonic Uncertainty Cascade** to quantify and propagate diagnostic uncertainty, cross-referencing patient history and medical literature to produce auditable, evidence-backed diagnoses on a single 24GB GPU.

YOLOv11n with Dendritic Optimization | *Python, PyTorch, Ultralytics, PerforatedAI*

[GitHub](#)

- Integrated PerforatedAI dendritic optimization into a custom YOLOv11n/Ultralytics training loop with adaptive plateau detection for optimal dendrite addition.
- Improved detection accuracy by **+2.30 mAP50** (100% data) and **+2.90 mAP50** (50% data) over baseline; work was adopted as an official base example in PerforatedAI's open-source repository.

VesselWatch — Oil Spill Detection System | *Python, Apache Kafka, Apache Airflow, Sentinel-1 SAR*

[GitHub](#)

- Built a production-grade, event-driven pipeline that fuses live **AIS** vessel-traffic streams with satellite **Synthetic Aperture Radar (SAR)** imagery to detect and validate potential oil spills.
- Orchestrated a deep-learning anomaly detector with **Kafka** and **Airflow** to auto-trigger Sentinel-1 satellite validation, enabling rapid, all-weather maritime environmental response.

ACHIEVEMENTS

- OffPay (Offline UPI Payments)**: Drove **200K+ impressions**, **1,000+ upvotes**, and **420+ downloads** from a community product launch.
- Finalist, Meta PyTorch OpenEnv Hackathon** — selected among **800 finalist teams** from **31,000+ registrations** (Meta · PyTorch · Hugging Face), for a PII-redaction environment.
- 1st Place / 957 participants** — Student Innovators Without Borders Hacks, for *SafeWander*, a dementia-patient monitoring system.
- 1st Place / 693 participants** — International Hackathon by Perforated AI; submission adopted as an official base example in PerforatedAI's open-source repository.
- Lemma AI Hackathon**: Secured **two winning positions (6th and 11th)** out of **450 submissions**, among 11 total winners.

TECHNICAL SKILLS

Languages: Python, TypeScript, JavaScript, Go, Rust, C++, Java, SQL

AI / ML: PyTorch, LangGraph, LangChain, RAG, LLMs, Multi-Agent Systems, Reinforcement Learning (PPO), Computer Vision, YOLO/Ultralytics, Hugging Face Transformers

Backend & Data: FastAPI, Node.js, Apache Kafka, Apache Airflow, Redis, PostgreSQL, Qdrant, SQLAlchemy

Web & DevOps: React, Next.js, Tailwind CSS, Docker, Kubernetes, AWS (EC2), Git, Linux, CI/CD